In [1]:
```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sqlalchemy import create_engine
import warnings
from datetime import datetime

warnings.filterwarnings('ignore')
```

In [2]:
```
username = "root"
password = "*******"
host = "localhost"
port = "3306"
database = "superstore_db"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}"
)

df = pd.read_sql("SELECT * FROM superstore_campaign", engine)

df.head()
```

Out[2]:

	Id	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer
0	0	1985	Graduation	Married	70951.0	0	0	2013-04-05
1	1	1961	Graduation	Single	57091.0	0	0	2014-06-15
2	9	1975	Master	Single	46098.0	1	1	2012-08-18
3	13	1947	PhD	Widowed	25358.0	0	1	2013-07-22
4	17	1971	PhD	Married	60491.0	0	1	2013-06-09

5 rows × 25 columns

In [3]:
```
df.shape
```

Out[3]:
```
(2240, 25)
```

In [4]:
```
df.dtypes
```

Out[4]:
```
Id                     int64
Year_Birth             int64
Education             object
Marital_Status        object
Income               float64
Kidhome                int64
Teenhome               int64
Dt_Customer           object
Recency                int64
MntWines               int64
MntFruits              int64
MntMeatProducts        int64
MntFishProducts        int64
MntSweetProducts       int64
MntGoldProducts        int64
NumDealsPurchases      int64
NumWebPurchases        int64
NumCatalogPurchases    int64
NumStorePurchases      int64
NumWebVisitsMonth      int64
Response               int64
Complain               int64
Age                    int64
Total_Spend            int64
Total_Purchases        int64
dtype: object
```

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
Id                     0
Year_Birth             0
Education              0
Marital_Status         0
Income                24
Kidhome                0
Teenhome               0
Dt_Customer            0
Recency                0
MntWines               0
MntFruits              0
MntMeatProducts        0
MntFishProducts        0
MntSweetProducts       0
MntGoldProducts        0
NumDealsPurchases      0
NumWebPurchases        0
NumCatalogPurchases    0
NumStorePurchases      0
NumWebVisitsMonth      0
Response               0
Complain               0
Age                    0
Total_Spend            0
Total_Purchases        0
dtype: int64
```

In [6]:
```
df.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

In [7]:
```
df['Education'].nunique()
```

Out[7]:
```
4
```

In [8]:
```
df['Total_Spend'].sum()
```

Out[8]:
```
np.int64(1356988)
```

Data Cleaning

In [9]:
```
df['Income'].fillna(df['Income'].median(), inplace=True)
```

In [10]:
```
df.head()
```

Out[10]:

	Id	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer
0	0	1985	Graduation	Married	70951.0	0	0	2013-04-05
1	1	1961	Graduation	Single	57091.0	0	0	2014-06-15
2	9	1975	Master	Single	46098.0	1	1	2012-08-18
3	13	1947	PhD	Widowed	25358.0	0	1	2013-07-22
4	17	1971	PhD	Married	60491.0	0	1	2013-06-09

5 rows × 25 columns

In [11]:
```
# Spending columns
spend_cols = [
    'MntWines','MntFruits','MntMeatProducts',
    'MntFishProducts','MntSweetProducts','MntGoldProducts'
]

df[spend_cols] = df[spend_cols].fillna(0)
```

Feature Engineering:

In [12]:
```
# Compute Age from Year_Birth.
current_year = datetime.now().year
df['Age'] = current_year - df['Year_Birth']
df[['Year_Birth','Age']].head(10)
```

Out[12]:

	Year_Birth	Age
0	1985	41
1	1961	65
2	1975	51
3	1947	79
4	1971	55
5	1965	61
6	1976	50
7	1960	66
8	1958	68
9	1987	39

In [13]:

```
# Calculate customer tenure from Dt_Customer.

df['Dt_Customer'] = pd.to_datetime(df['Dt_Customer'])
df['Tenure'] = (datetime.now() - df['Dt_Customer']).dt.days / 365
```

In [14]:

```
# Compute Total Spend by summing all Mnt* columns.
df['Total_Spend'] = df[spend_cols].sum(axis=1)
```

In [15]:

```
# Determine Category_Breadth by counting how many product categories have non-zero
df['Category_Breadth'] = (df[spend_cols] > 0).sum(axis=1)
df[['Category_Breadth','Id']].head()
```

Out[15]:

	Category_Breadth	Id
0	6	0
1	5	1
2	3	9
3	3	13
4	6	17

In [16]:

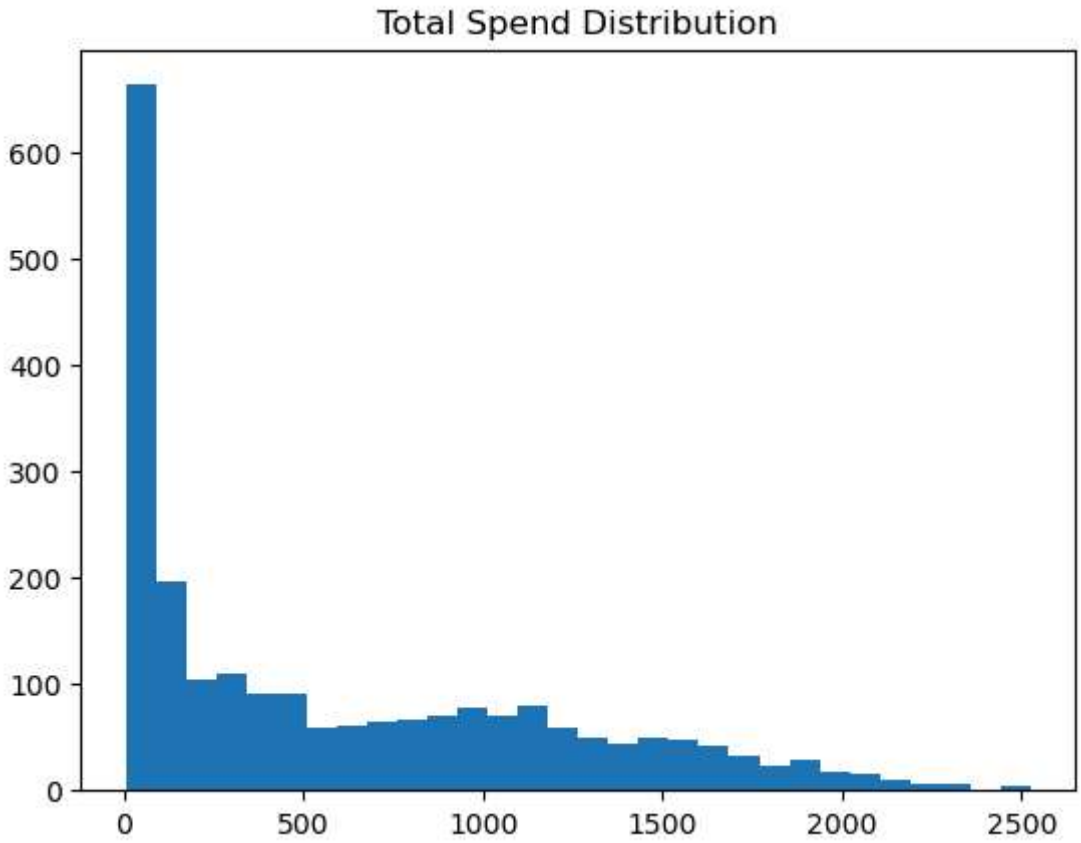
```
# Derive engagement metrics from the Num* columns, such as total purchases or ratio
df['Total_Purchases'] = (
    df['NumWebPurchases'] +
    df['NumCatalogPurchases'] +
    df['NumStorePurchases']
)
```

In [17]:

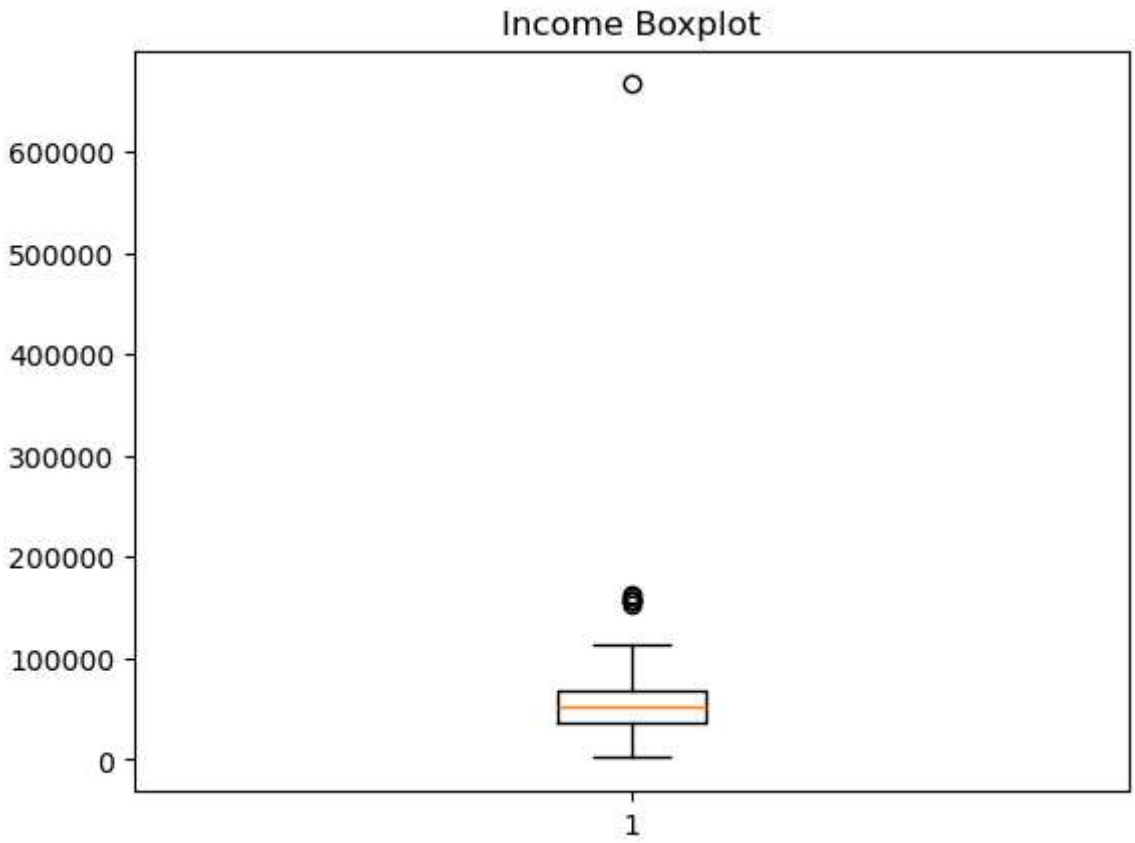
```
# web purchase ratio
df['Web_Ratio'] = df['NumWebPurchases'] / df['Total_Purchases']
```

Exploratory Data Analysis (EDA)

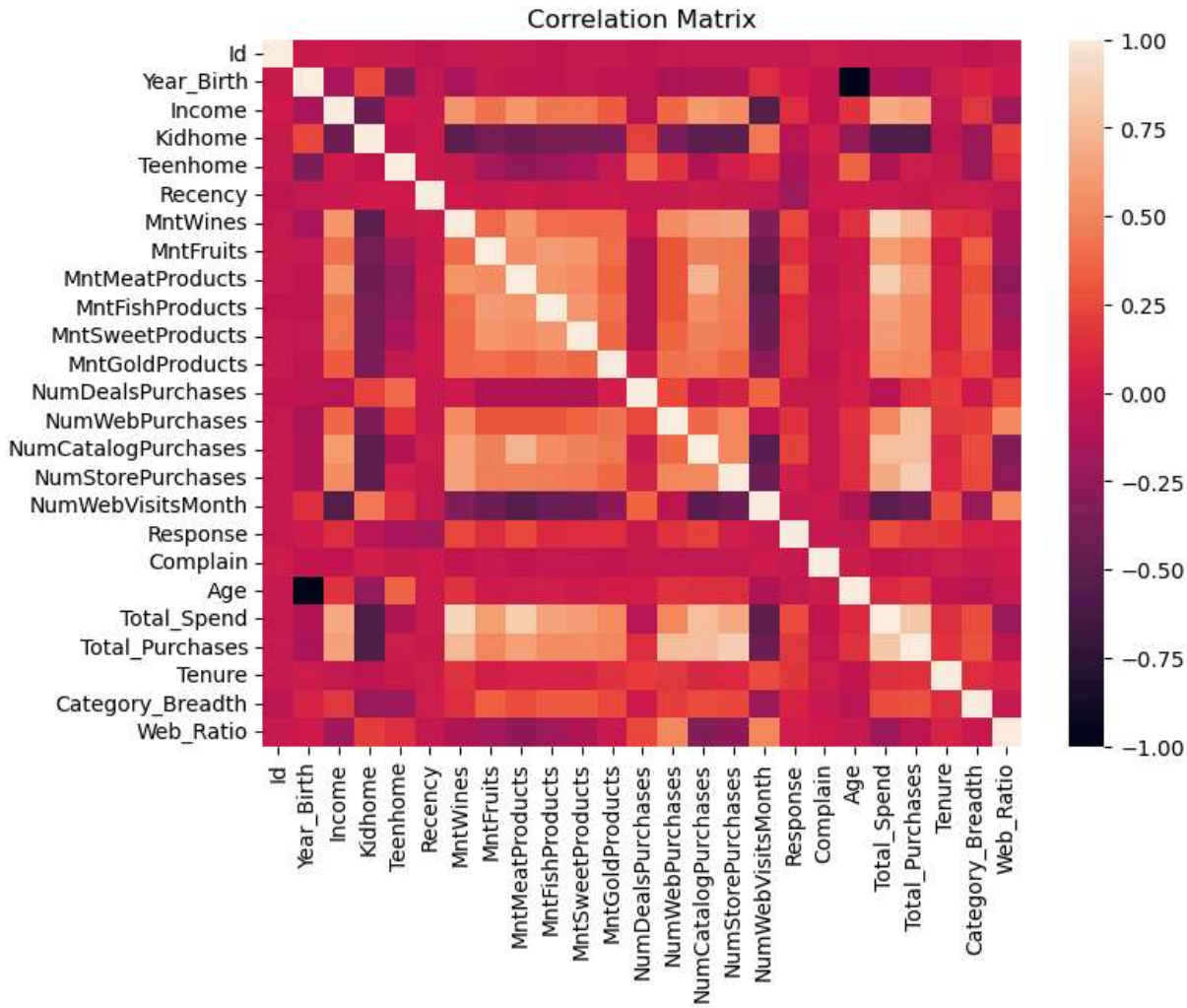
In [18]:
```
# Histogram (Total Spend)
plt.hist(df['Total_Spend'], bins=30)
plt.title("Total Spend Distribution")
plt.show()
```



In [19]:
```
# Boxplot (Income)

plt.boxplot(df['Income'])
plt.title("Income Boxplot")
plt.show()
```



In [20]:
```
numeric_df = df.select_dtypes(include=['number'])

plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=False)
plt.title("Correlation Matrix")
plt.show()
```



In [21]:
```
# High Spending Segment
high_spenders = df[df['Total_Spend'] > df['Total_Spend'].median()]
high_spenders.shape
```

Out[21]: (1119, 28)

In [22]:
```
# Under-Penetrated Segment
df[(df['Income'] > df['Income'].median()) &
   (df['Total_Spend'] < df['Total_Spend'].median())]
```

Out[22]:

	Id	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Cus
10	48	1964	Graduation	Married	55761.0	0	1	2014
14	73	1953	PhD	Single	51411.0	1	2	2013
23	125	1958	Basic	Married	53083.0	1	1	2013
30	176	1980	Basic	Single	67506.0	0	1	2014
48	246	1965	Graduation	Married	66480.0	1	1	2014
...	...	...	...	...	...	...	...	
2205	11004	1893	Basic	Single	60182.0	0	1	2014
2212	11039	1975	Graduation	Married	51948.0	0	1	2013
2223	11096	1952	Basic	Married	57247.0	0	1	2013
2224	11100	1972	Graduation	Divorced	51813.0	1	1	2013
2236	11181	1949	PhD	Married	156924.0	0	0	2013

123 rows × 28 columns

In [23]:
```
df.head()
```

Out[23]:

	Id	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer
0	0	1985	Graduation	Married	70951.0	0	0	2013-04-05
1	1	1961	Graduation	Single	57091.0	0	0	2014-06-15
2	9	1975	Master	Single	46098.0	1	1	2012-08-18
3	13	1947	PhD	Widowed	25358.0	0	1	2013-07-22
4	17	1971	PhD	Married	60491.0	0	1	2013-06-09

5 rows × 28 columns

Statistical Analysis

In [24]:
```
#Group comparisons
df.groupby('Kidhome')['Total_Spend'].mean()
```

Out[24]:
```
Kidhome
0    901.260634
1    205.434928
2    145.250000
Name: Total_Spend, dtype: float64
```

In [25]:
```
from scipy.stats import ttest_ind

with_kids = df[(df['Kidhome'] > 0) | (df['Teenhome'] > 0)]['Total_Spend']
no_kids = df[(df['Kidhome'] == 0) & (df['Teenhome'] == 0)]['Total_Spend']

t_stat, p_value = ttest_ind(with_kids, no_kids)

print("Average Spend (With Kids):", with_kids.mean())
print("Average Spend (No Kids):", no_kids.mean())
print("P-value:", p_value)
```

```
Average Spend (With Kids): 406.57990012484396
Average Spend (No Kids): 1106.0297805642633
P-value: 2.023752068671191e-158
```

In [26]:
```
# Acceptance Rate Comparison
with_kids_resp = df[(df['Kidhome'] > 0) | (df['Teenhome'] > 0)]['Response']
no_kids_resp = df[(df['Kidhome'] == 0) & (df['Teenhome'] == 0)]['Response']

t_stat_resp, p_value_resp = ttest_ind(with_kids_resp, no_kids_resp)

print("Acceptance Rate (With Kids):", with_kids_resp.mean())
print("Acceptance Rate (No Kids):", no_kids_resp.mean())
print("P-value (Acceptance):", p_value_resp)
```

```
Acceptance Rate (With Kids): 0.10299625468164794
Acceptance Rate (No Kids): 0.26489028213166144
P-value (Acceptance): 1.0424367047639332e-22
```

Variance Analysis (ANOVA)

In [27]:
```
#Education vs Total Spend (ANOVA)
df.groupby('Education')['Total_Spend'].mean()
```

Out[27]:
```
Education
Basic         409.385214
Graduation    619.898846
Master        611.781081
PhD           672.409465
Name: Total_Spend, dtype: float64
```

In [28]:
```
from scipy.stats import f_oneway

groups = [group['Total_Spend'].values
          for name, group in df.groupby('Education')]

f_stat, p_value = f_oneway(*groups)

print("ANOVA p-value:", p_value)
```

```
ANOVA p-value: 1.8361393210251286e-07
```

In [29]:
```
# Relationship test (Chi-square)

# Chi-Square Test (Education vs Response)
```

```
from scipy.stats import chi2_contingency

table = pd.crosstab(df['Education'], df['Response'])

chi2, p, dof, expected = chi2_contingency(table)

print("Chi-square p-value:", p)
```

```
Chi-square p-value: 8.840050548990529e-05
```

In [30]:
```
# Marital Status vs Response
table_marital = pd.crosstab(df['Marital_Status'], df['Response'])

chi2, p, dof, expected = chi2_contingency(table_marital)

print("Chi-square p-value (Marital):", p)
```

```
Chi-square p-value (Marital): 6.164467480815522e-11
```

In [31]:
```
# Income Bracket ANOVA
df['Income_Bracket'] = pd.qcut(df['Income'], 3,
                               labels=['Low','Medium','High'])

groups_income = [group['Total_Spend'].values
                 for name, group in df.groupby('Income_Bracket')]

from scipy.stats import f_oneway
f_oneway(*groups_income)
```

Out[31]:
```
F_onewayResult(statistic=np.float64(2155.9017791524607), pvalue=np.float64(0.0))
```

Modelling

In [32]:
```
#Model Objective
#Predicting Gold Membership Acceptance (Response)

#The goal is to build a classification model to predict whether a customer will acc
```

In [33]:
```
#Feature Selection
# Selection of Key Predictive Variables
# Selecting important features
features = ['Income', 'Age', 'Total_Spend',
            'NumWebPurchases', 'Recency']
X = df[features]
y = df['Response']
```

In [34]:
```
# Data Splitting Strategy
# Train-Test Split (70% Training, 30% Testing)
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
```

In [35]:
```
# Model Development
# Random Forest Classification Model
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train, y_train)
```

Out[35]:
```
▾ RandomForestClassifier
▶ Parameters
```

In [36]:
```
# MODEL 1 — Original Random Forest

# from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, classification_report

y_pred = rf_model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy*100:.2f}%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))
```

```
Model Accuracy: 85.86%

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.96      0.92       575
           1       0.52      0.26      0.34        97

    accuracy                           0.86       672
   macro avg       0.70      0.61      0.63       672
weighted avg       0.83      0.86      0.84       672
```

In [37]:
```
# MODEL 2 — Class Weight Balanced

rf_balanced = RandomForestClassifier(class_weight='balanced', random_state=42)
rf_balanced.fit(X_train, y_train)
y_pred_balanced = rf_balanced.predict(X_test)

print("=== MODEL 2: Class Weight Balanced ===")
print(f"Accuracy: {accuracy_score(y_test, y_pred_balanced)*100:.2f}%")
print(classification_report(y_test, y_pred_balanced))
```

```
=== MODEL 2: Class Weight Balanced ===
Accuracy: 85.71%
              precision    recall  f1-score   support

           0       0.88      0.96      0.92       575
           1       0.51      0.23      0.31        97

    accuracy                           0.86       672
   macro avg       0.70      0.60      0.62       672
weighted avg       0.83      0.86      0.83       672
```

In [38]:

```python
# MODEL 3 — SMOTE (Best for Marketing)

from imblearn.over_sampling import SMOTE

smote = SMOTE(random_state=42)
X_train_smote, y_train_smote = smote.fit_resample(X_train, y_train)

print("=== Before SMOTE ===")
print(f"Class 0: {sum(y_train==0)}, Class 1: {sum(y_train==1)}")
print("=== After SMOTE ===")
print(f"Class 0: {sum(y_train_smote==0)}, Class 1: {sum(y_train_smote==1)}")

rf_smote = RandomForestClassifier(random_state=42)
rf_smote.fit(X_train_smote, y_train_smote)
y_pred_smote = rf_smote.predict(X_test)

print("\n=== MODEL 3: SMOTE ===")
print(f"Accuracy: {accuracy_score(y_test, y_pred_smote)*100:.2f}%")
print(classification_report(y_test, y_pred_smote))
```

```
=== Before SMOTE ===
Class 0: 1331, Class 1: 237
=== After SMOTE ===
Class 0: 1331, Class 1: 1331

=== MODEL 3: SMOTE ===
Accuracy: 77.23%
              precision    recall  f1-score   support

           0       0.90      0.83      0.86       575
           1       0.30      0.44      0.36        97

    accuracy                           0.77       672
   macro avg       0.60      0.64      0.61       672
weighted avg       0.81      0.77      0.79       672
```

In [39]:

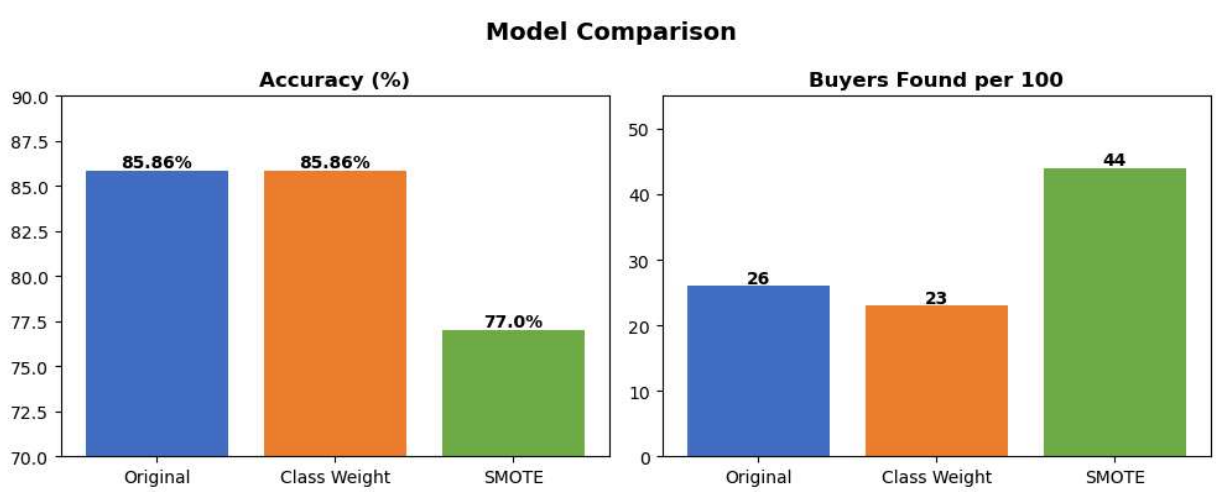
```python
# FINAL COMPARISON

print("=== FINAL COMPARISON ===")
print("Model 1 Original      — Buyers found: 26 out of 100")
print("Model 2 Class Weight  — Buyers found: 23 out of 100")
print("Model 3 SMOTE         — Buyers found: 44 out of 100")
```

```
print("\nBest for marketing = SMOTE")
print("Reason: Limited budget allows for more actual buyers")
```

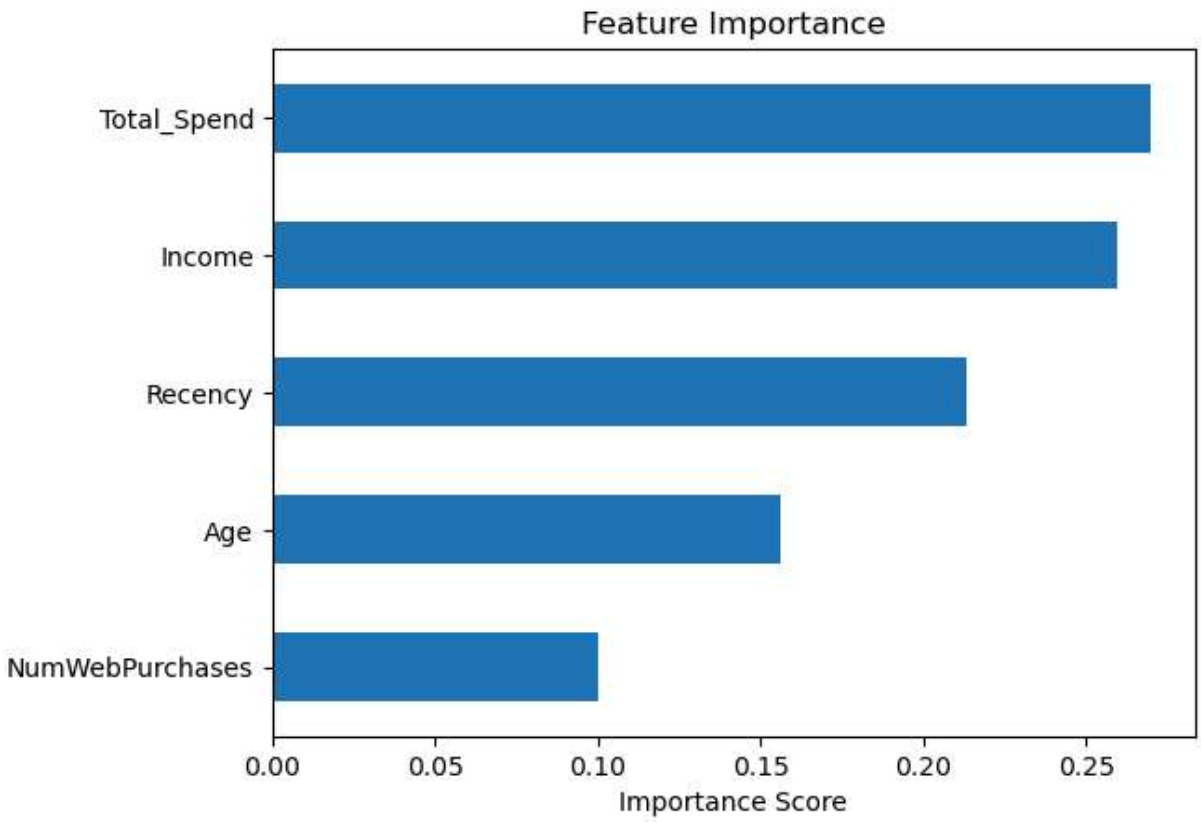
```
=== FINAL COMPARISON ===
Model 1 Original      — Buyers found: 26 out of 100
Model 2 Class Weight  — Buyers found: 23 out of 100
Model 3 SMOTE         — Buyers found: 44 out of 100

Best for marketing = SMOTE
Reason: Limited budget allows for more actual buyers
```

In [46]:

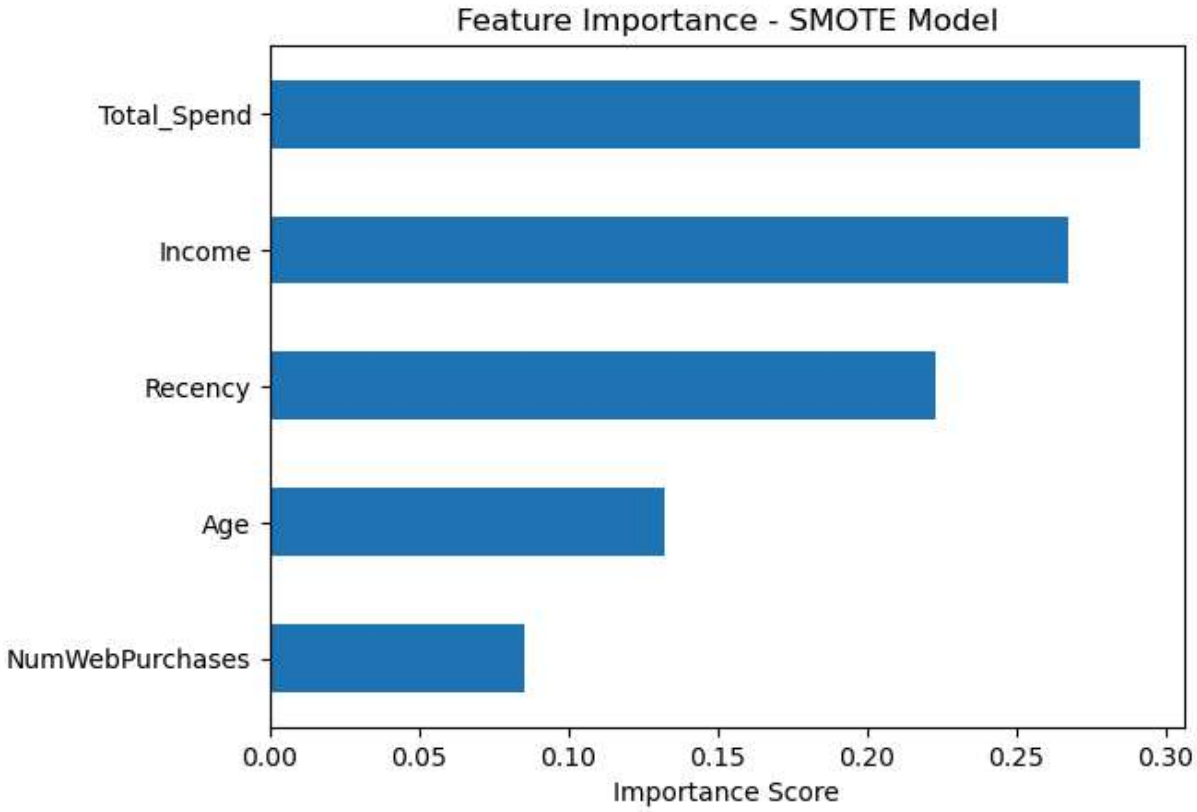
```
import matplotlib.pyplot as plt

# Data
models = ['Original', 'Class Weight', 'SMOTE']
accuracy = [85.86, 85.86, 77.0]
recall   = [26, 23, 44]
colors   = ['#4472C4', '#ED7D31', '#70AD47']

# Two charts side by side
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(10, 4))

# Chart 1 — Accuracy
ax1.bar(models, accuracy, color=colors)
ax1.set_title('Accuracy (%)', fontweight='bold')
ax1.set_ylim(70, 90)
for i, v in enumerate(accuracy):
    ax1.text(i, v + 0.2, f'{v}%', ha='center', fontweight='bold')

# Chart 2 — Buyers Found
ax2.bar(models, recall, color=colors)
ax2.set_title('Buyers Found per 100', fontweight='bold')
ax2.set_ylim(0, 55)
for i, v in enumerate(recall):
    ax2.text(i, v + 0.5, str(v), ha='center', fontweight='bold')

plt.suptitle('Model Comparison', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('model_comparison.png', dpi=150)
plt.show()
```



In [40]:
```
# Feature Importance Analysis
# Identifying Key Drivers of Membership Acceptance
import pandas as pd
import matplotlib.pyplot as plt

importance = pd.Series(
    rf_model.feature_importances_,
    index=features
).sort_values()

importance.plot(kind='barh')
plt.title('Feature Importance')
plt.xlabel('Importance Score')
plt.show()
```



In [41]:
```
# Feature Importance — SMOTE Model
importance_smote = pd.Series(
    rf_smote.feature_importances_,
    index=features
).sort_values()
importance_smote.plot(kind='barh')
plt.title('Feature Importance - SMOTE Model')
plt.xlabel('Importance Score')
plt.show()
```



In [47]:
```
df.to_sql(
    name='superstore_campaign_clean',
    con=engine,
    if_exists='append',
    index=False
)
```

Out[47]: 2240